

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION(3 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO1 – Manipulation (BL2)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	30 (3 Questions × 10 Marks)
CO1 Statement:	<i>Understand the fundamental concepts, principles, and techniques of data visualization (BL2).</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 3 questions, each carrying 10 mark. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Mapping Data onto Aesthetics	Mapping categorical, ordinal, quantitative data to position, size, color	1	10
Coordinate Systems and Axes	Cartesian vs Polar systems, axis design	2	10
Color Scales & Usage	Color to distinguish, represent values,	3	10
GRAND TOTAL:			30 Marks

1.A city transportation department collects data on different modes of transport, passenger satisfaction levels, and average daily ridership.

Transport Mode	Satisfaction Level	Daily Ridership
Bus	High	12,000
Metro	Very High	25,000
Auto Rickshaw	Medium	8,000
Taxi	Low	4,500
Bicycle Sharing	High	6,500

Design a visualization by mapping categorical, ordinal, and quantitative data onto suitable aesthetics such as position, shape, and size. Select appropriate scales for each variable and justify how your design enables effective comparison of transportation modes and user satisfaction.

2.A university library tracks the number of books borrowed from different sections.

Section	Books Borrowed
Science	850
Literature	620
Technology	1100
History	480
Management	730

Design a visualization that uses color to distinguish library sections and highlight the section with the highest borrowing rate. Choose an effective and accessible color palette, explaining how color is used both to distinguish categories and to emphasize the most utilized section while maintaining readability.

3.A financial analyst tracks the monthly returns of different company sectors.

Sector	Monthly Return (%)
Information Technology	12
Banking	8
Healthcare	15
Manufacturing	5
Energy	-3

A visualization designer must decide whether to use a linear or nonlinear axis to represent the monthly returns. Design an appropriate visualization and justify the choice of coordinate system and axis scale. Explain how the selected design helps users accurately compare sector performance, identify outliers, and interpret both positive and negative returns effectively.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Technical Content Accuracy	30%	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood		
Problem Identification	25%	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach		
Use of Tools	20%	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals		
Communication	15%	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides		
Professionalism	10%	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____